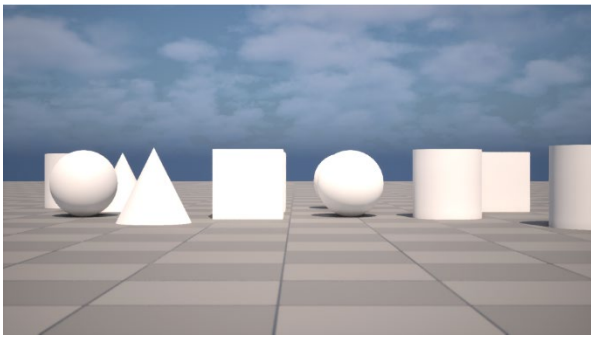
UU Camera User Guide

Version 1.0 – December 2025

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UU Camera is a custom camera actor plugin designed for video production, virtual production where you need to be able to create a Garbage Mask on one or more parts of the scene. It extends the functionality of Unreal Engine's CineCamera to provide a clean and controllable alpha output based on stencil masks, ready to be sent to a hardware output (AJA/Blackmagic).





Overview

The UltiUnreal camera allows you to use a CineCamera in -game mode, launched via a blueprint, without going through a Viewport capture.

Drop BP_UUCameraOutput.on your level, assign a stencil value of 1 to the objects to appear in the alpha image and UU Camera generates an alpha channel appropriate for external keyers, such as Blackmagic Ultimatte or Vmix. We are doing external compositing here.

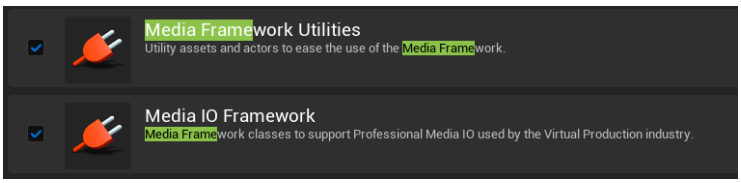
The plugin is particularly suitable for any workflow where Unreal needs to produce a clean RGB+Alpha signal.

Installation

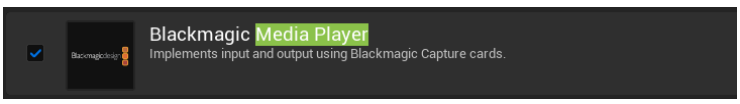
UU Camera comes as a standard plugin for Unreal Engine. It was developed and tested with Unreal Engine 5.7.



Go to the plugins window and enable UltiUnrealCamera and CineCameraSceneCapture



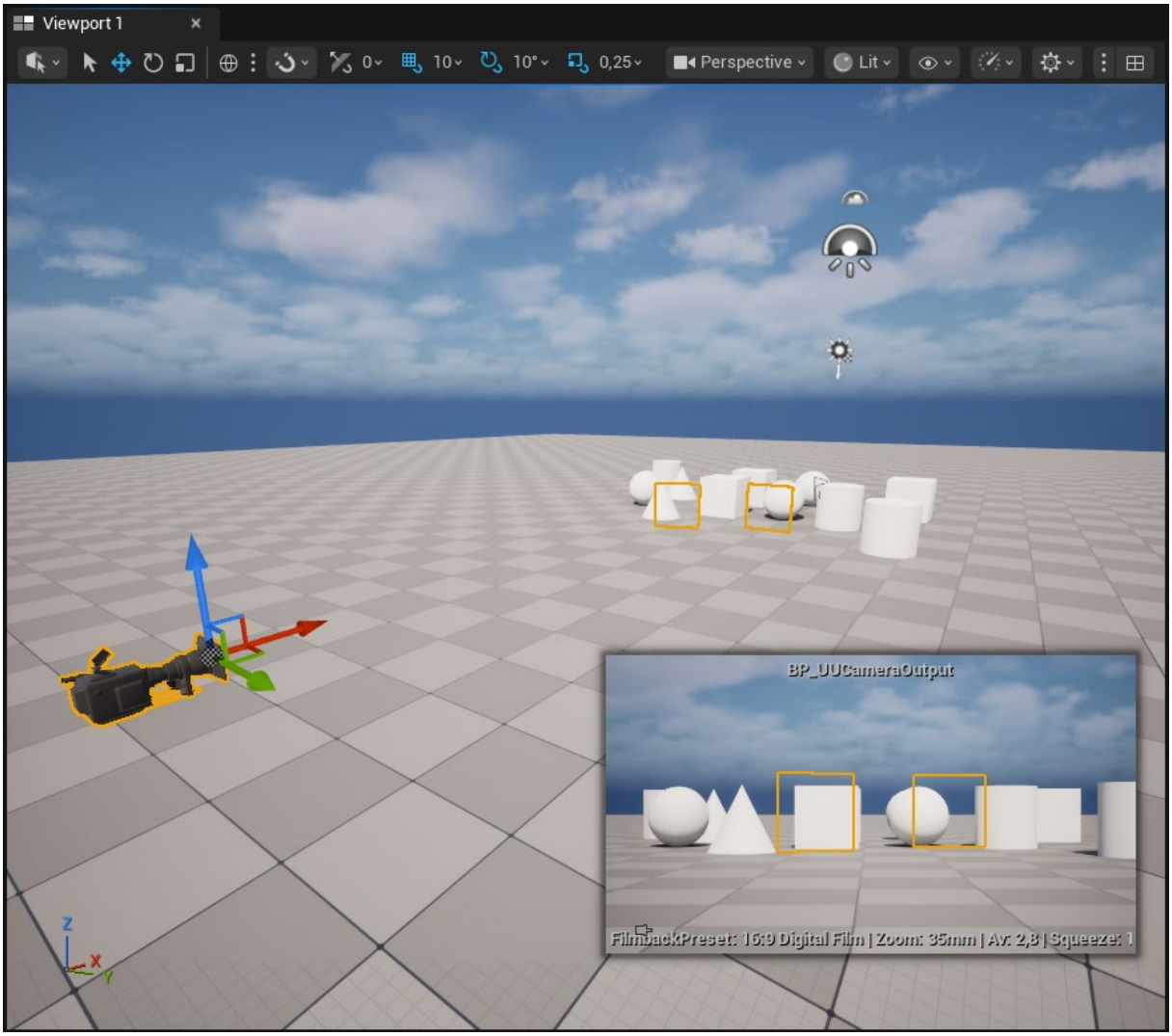
Also enable "Media Framework Utilities" and "Media IO Framework"



You also need to activate your video card plugin "Blackmagix Media Player" or "Aja Media Player"
Then restart your project.

Project Setup – Garbage Alpha

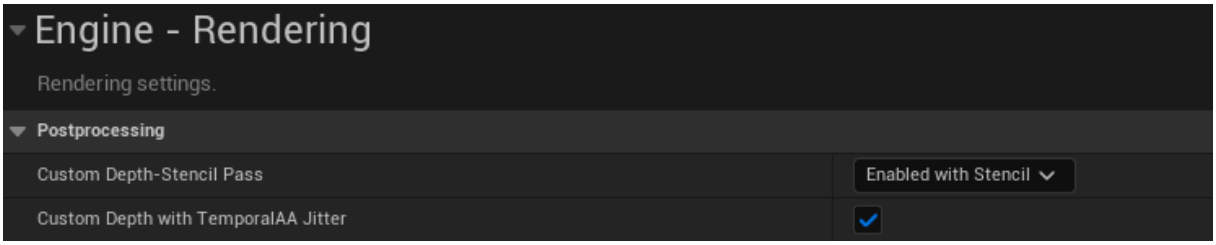
UU Camera uses custom stencil values to decide which objects are written to the alpha channel.



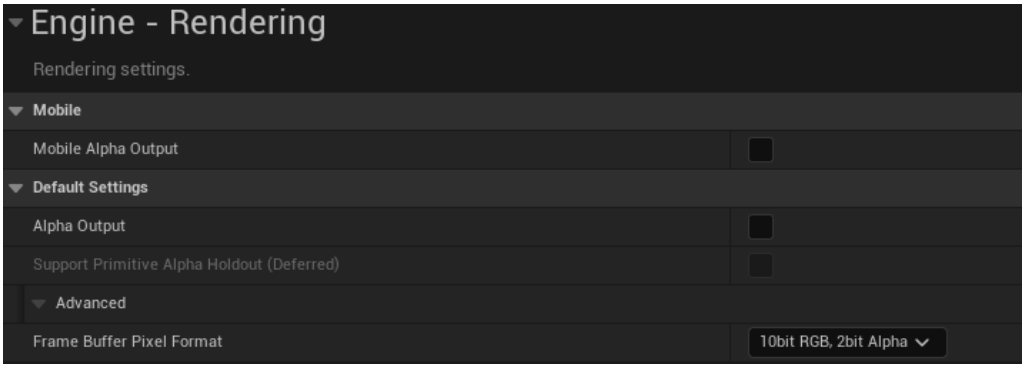
In the image above, you can see two planes circled in yellow in front of the white objects. These are the masks that we want to bring out on the alpha channel.

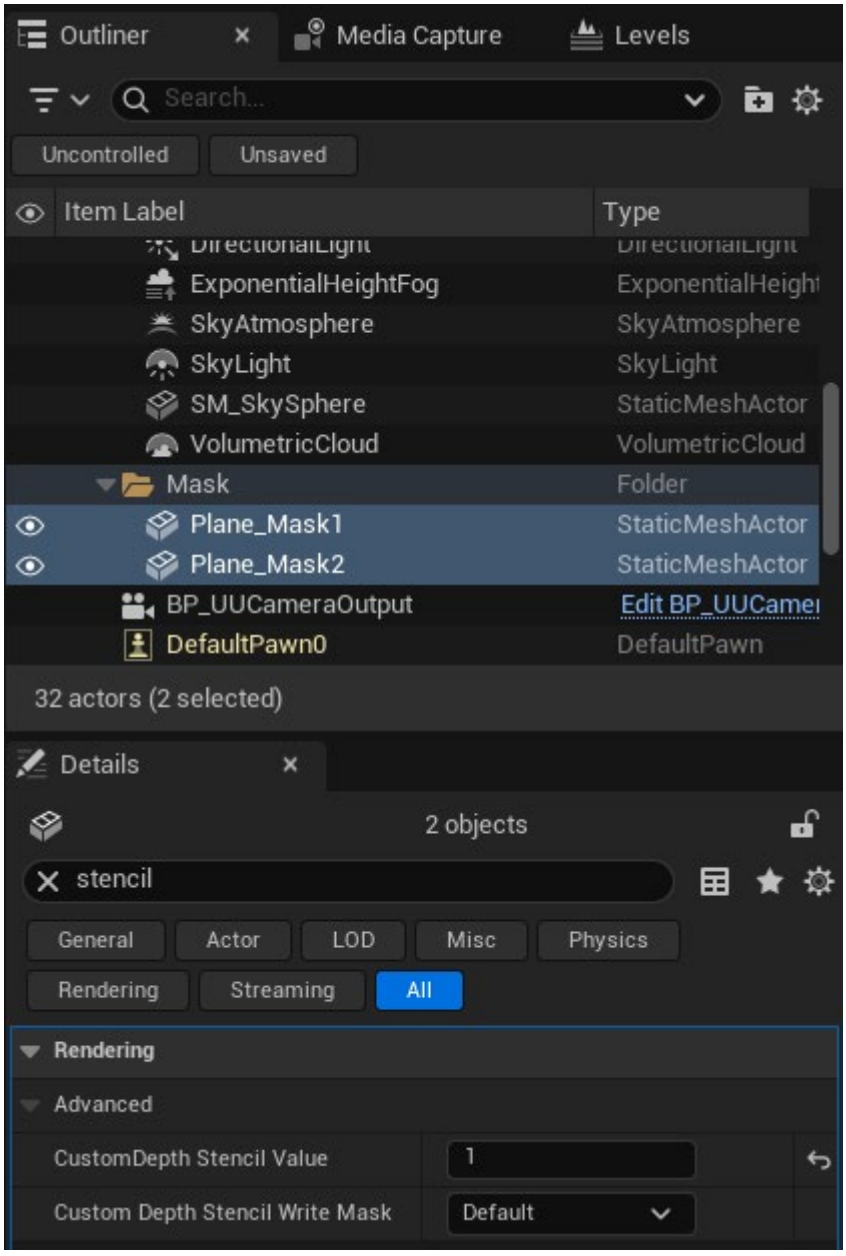
Before that, you need to enable the project-level tencil:

- Open the project settings in Unreal.
- Search for " **Custom Depth-Stencil Pass**".
- Set it to " **Enable with Stencil**".



Alpha Output is not mandatory.





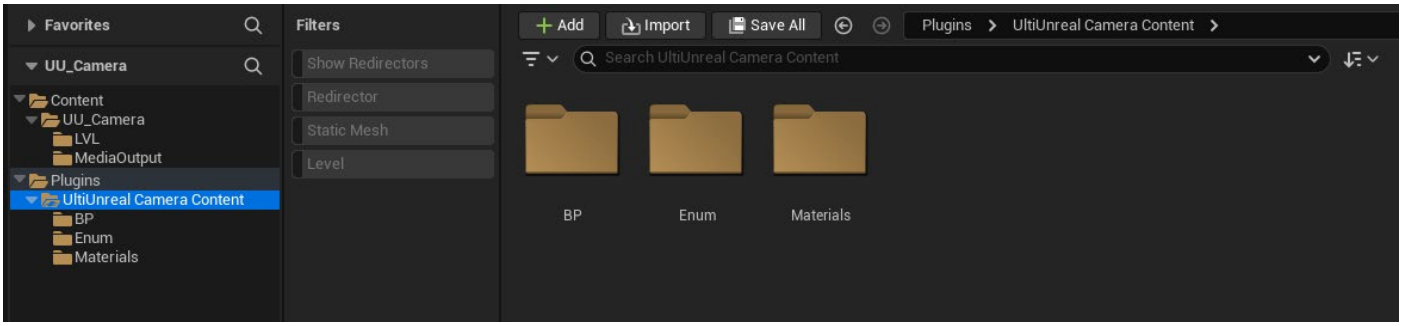
Next, select the objects that should appear on the alpha channel (talent, props, CG scenery, AR graphics, etc.) and assign it a stencil value as described in the image above:

- Select it or the objects in the level



- In the Details panel, turn on "Render CustomDepth Pass."
- Set "CustomDepth Stencil Value" to 1.

In the plugin files, you have a Material called M_hide_Mask in Metarial Folder. Apply Material to objects to make them invisible and counted by the camera as objects like Garbage.

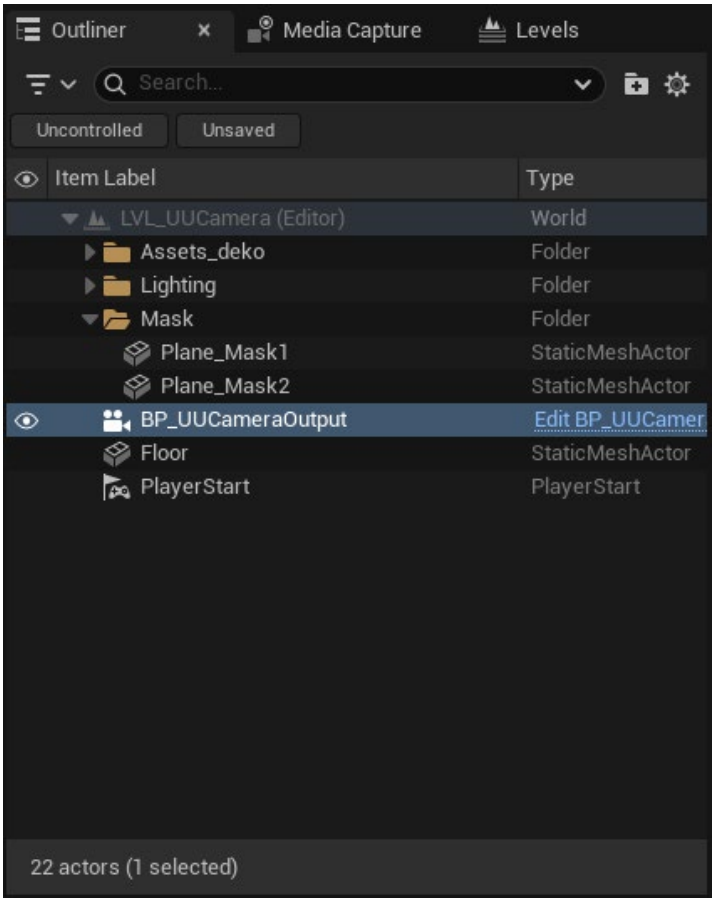


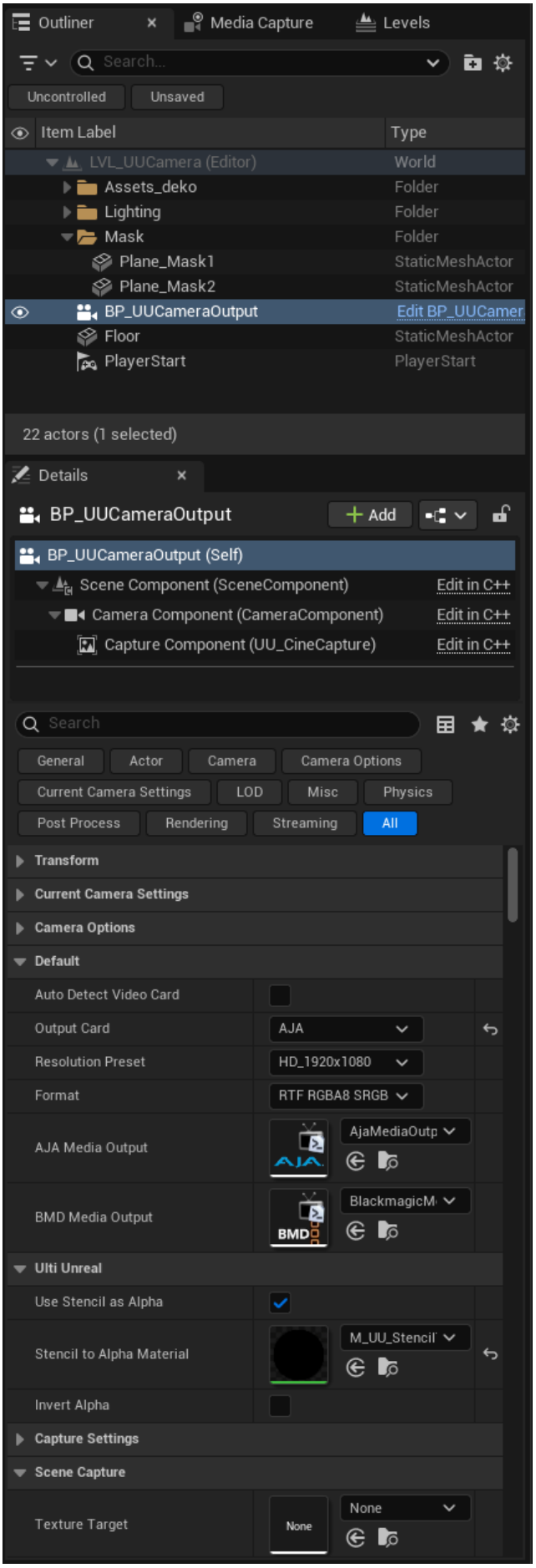
Using BP_UUCameraOutput



Once the plugin is activated and the stencil is configured, you can start using the UU Camera actor in your level:

- In the Place Acteurs panel, look for BP_UUCameraOutput. You can also drag the contained blueprint into the BP plugin folder.
- Drag BP_UUCameraOutput into your level.
- Position the camera and orient it.
- In the Details panel, configure the "Default" and "UltiUnreal" categories





The plugin automatically inserts the Material M_UU_StencilToAlpha into Stencil To Alpha Material.

Setup

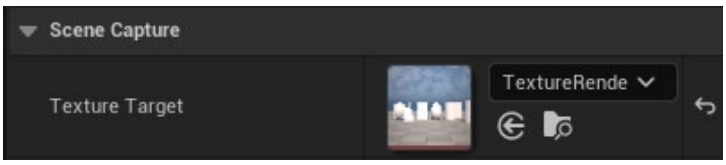
The controls in the UltiUnreal category are:

- **Use Stencil As Alpha**: Allows stencil based alpha generation.
- **Invert Alpha**:– Reverses the alpha to be able to process Garbage objects.

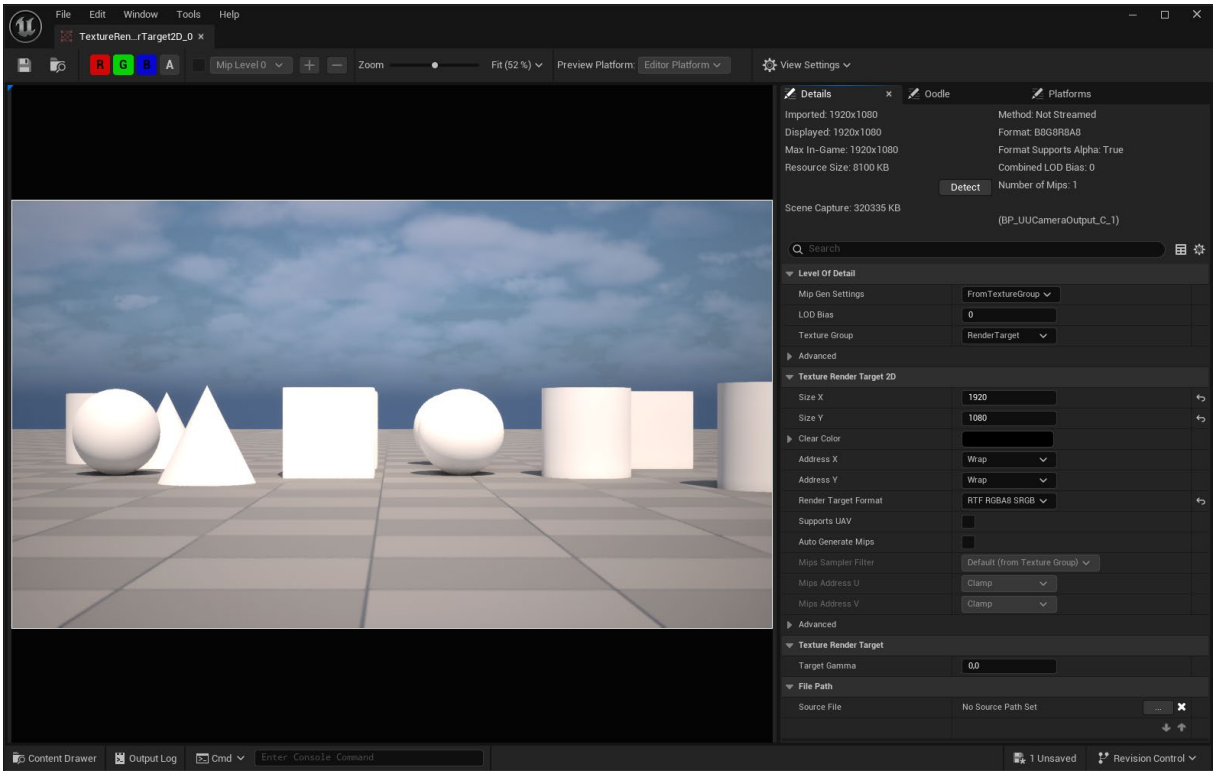
The controls in the Default base category are:

- **Auto Detect Video Card**: When you launch your project in -game mode, add a "-aja" or "-blackmagic" argument so that the plugin automatically assigns the correct card setting. This is useful in case of multi-machines and synchronized projects.
- **Output Card**: If you do not choose the option to auto-detect the video card, select the card on the computer.
- **Resolution Preset**: This is the resolution used by the Render Target to write the video frame. Be careful, this is not the output configuration of your Aja or Blackmagic card.
- Format: This is the type of writing of the Render Target. Since you output two full images: one color and the other in black and white (no gray), you can simply choose "RTF RGBA8"
- **AJA Media Output**: Place your configured AJA Media Output Asset here.
- **BMD Media Output** : Place your configured BlackMagic Media Output asset here when using a Decklink card.

During PLAY, the asset in the Scene Capture category, Texture Target is automatically generated and applied. This is your Render Target.

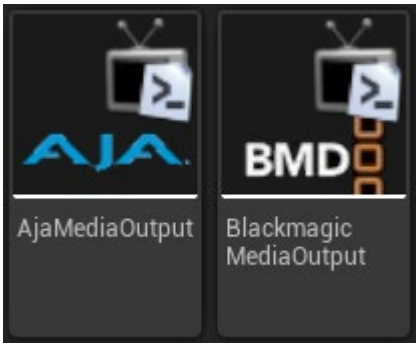


You can also configure all the standard settings of the cinema camera (focal length, aperture, sensor size, focus, etc.), as the UU camera is built directly on ACineCameraActor.



In the image above, you can see the Render Target automatically applied during PLAY in the Scene Capture, Texture Target category

External Output (AJA / Blackmagic)



UU Camera is designed to work with Unreal's built-in Media Output/Media Capture pipeline. The typical configuration is as follows:

- Enable your AJA or Blackmagic plugin in the project settings.
- Create a Media Output asset (for example, an AJA Media Output).
- Configure your Media Output asset by selecting Alpha output (2 SDI: 1 RGB + 1 Alpha)
- Assign the asset to the Default category of the UU_Camera.

Once configured, your external switcher or keyer receives an RGB+Alpha signal from the UU camera, ready for compositing.



Example of how to configure a video card in the Media Output (Here AJA Kona)

If you encounter any unexpected artifacts or behaviors, try turning off post-processing volumes step-by-step (bloom, vignette, chromatic aberration, etc.) to identify which effects interact with your alpha output.

Support & Updates

UU Camera is part of the UltiUnreal toolset developed by JDarrasVP for multi-station virtual production workflows. Future updates may add new options, presets, or integration aids for specific broadcast environments.

Do not hesitate to ask your questions on my Youtube page at http://youtube.com/jdarrasvp or on my website: www.jdarrasvp.fr